

Capture-mark-recapture methods for abundance estimation



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Most methods differ in how they estimate p

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- And vice versa

$n = 5$ individuals captured over 3 sampling occasions

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Animal 1	0	0	1
Animal 2	1	1	0
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These data tell us about p and hence N . Estimation is usually achieved using maximum likelihood methods.

LINCOLN-PETERSON METHOD

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Later it was used by Lincoln (shown above) and Peterson to estimate fish and wildlife populations

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- On the first, n_1 animals are captured and marked



LINCOLN-PETERSON STUDY DESIGN

- There are only 2 capture occasions
- On the first, n_1 animals are captured and marked
- On the second, n_2 animals are captured and m_2 of them are recaptures



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And so . . .

$$\hat{N} = \frac{n_1 n_2}{m_2}$$

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- (3) No tag loss or mis-identification

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- Individual heterogeneity
- Combinations of the above

Model	Description
M_0	The most basic model in which p and c are constant
M_t	p differs among sampling occasions and $p_t = c_t$.
M_b	Behavioral response model in which p and c differ. Can describe trap happiness or trap shiness.
M_{tb}	A combination of models M_t and M_b .

where

- p = capture probability
- p_t = capture probability on occasion t
- c = recapture probability

REMOVAL SAMPLING



<https://youtu.be/W3lcHdFyZrQ>

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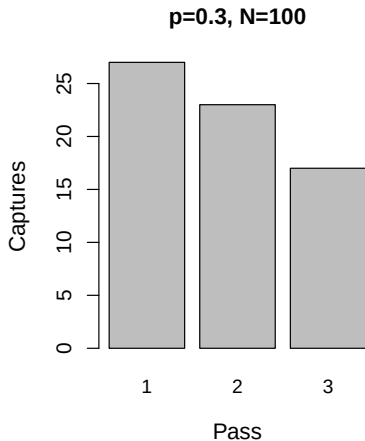
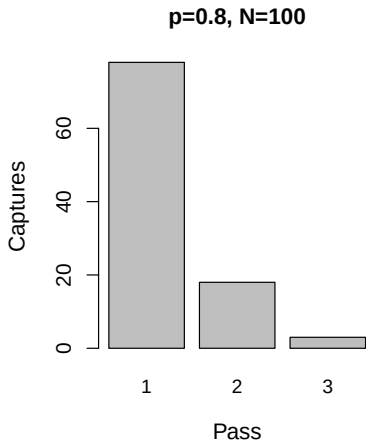
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The number of captures you would expect on each pass:

Pass	Expected captures
1	pN
2	$p(1 - p)N$
3	$p(1 - p)^2N$
4	$p(1 - p)^3N$
$\vdots$	$\vdots$
K	$p(1 - p)^{K-1}N$

EXAMPLE

The rate at which the population is depleted tells us about p , and therefore how many individuals we missed.



Key points

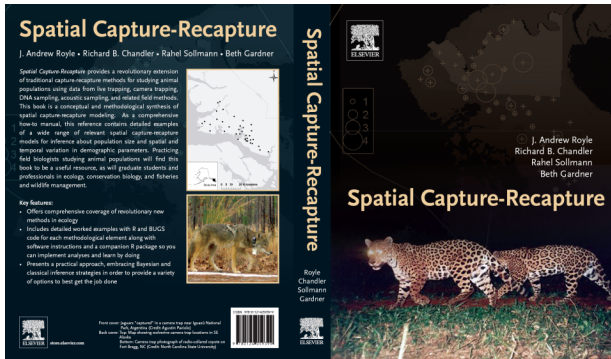
- Capture-recapture methods use information about recapture rates to estimate capture probability and abundance
- More advanced methods can be used to estimate density and vital rates
- Modern field methods use camera traps or DNA sampling techniques to collect non-invasive capture-recapture data

Lingering questions

- How do we convert abundance to density?
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Read Chapter 11